

NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI

MOBILE APPLICATION DEVELOPMENT LABORATORY

Placement Companion

An Android app for organized, personalized placement preparation

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PROJECT IMPLEMENTATION • GROUP PRESENTATION • INDIVIDUAL VIVA

From scattered preparation to a daily routine

01 / CONTEXT

The student problem

Company information, practice questions and notes are scattered.

Students struggle to prioritize weak topics and maintain a study schedule.

02 / GOALS

Our objectives

Organize preparation in one Android app.

Track tasks and retain study material locally.

Personalize guidance using AI.

Target users: students preparing for campus placements and interviews

Career readiness creates a practical student need

NEED

Career readiness

Placement preparation needs technical practice, interview confidence and a consistent routine.

CONTEXT

Personal guidance

AICTE NEAT supports EdTech that addresses learning gaps and employability skills.

PROPOSED CHANNEL

Campus access

Start with placement-season students; reach them through peer groups and placement cells.

Positioning: a daily preparation companion alongside practice and recruitment platforms.

Context: AICTE NEAT; NIT Trichy Training & Placement. Targeting and positioning are project proposals.

Student value must translate into repeat usage

Value & retention

One place for tasks, notes and AI help.

Saved progress can support a daily routine.

Adoption channels

Pilot with student groups and placement cells.

Use feedback to improve the preparation flow.

Possible business model

Free core planning; optional paid AI features.

Explore institutional plans after a campus pilot.

Costs & success measures

AI usage, support and future hosting costs.

Track retention, task completion and cost per user.

Proposed roadmap: pricing, subscriptions, institutional plans and analytics are not implemented.

One app, six connected preparation modules

01

Personalization

Target role, subjects and daily study hours.

02

Study planning

Tasks, timetable and completion tracking.

03

Company preparation

Companies, PYQs and interview experiences.

04

Study material

Notes, resources, search and bookmarks.

05

AI assistance

Roadmaps, doubt answers and interview feedback.

06

Daily navigation

Home, calendar and profile access.

What each technology contributes to the app

LAYER	TECHNOLOGY / TOOL	ROLE IN THIS PROJECT
Language & build	Kotlin · Android Studio · Gradle	Native Android code, builds and APK generation
UI & navigation	Compose · Material 3 · Navigation	Declarative screens, controls and route handling
State & async work	ViewModel · Coroutines · StateFlow	Screen state, background calls and reactive updates
Local persistence	Room / DAO · DataStore	8 structured entities; preferences and onboarding
Networking & AI	Retrofit · OkHttp · Gson · Groq	HTTP requests, JSON parsing and AI responses
Verification	Compose UI tests · AndroidJUnit4	Navigation checks and opt-in live API tests

Android compatibility: min SDK 26 (Android 8.0+) • compile / target SDK 35

UI actions pass through a layered MVVM design

SEPARATION OF CONCERNS

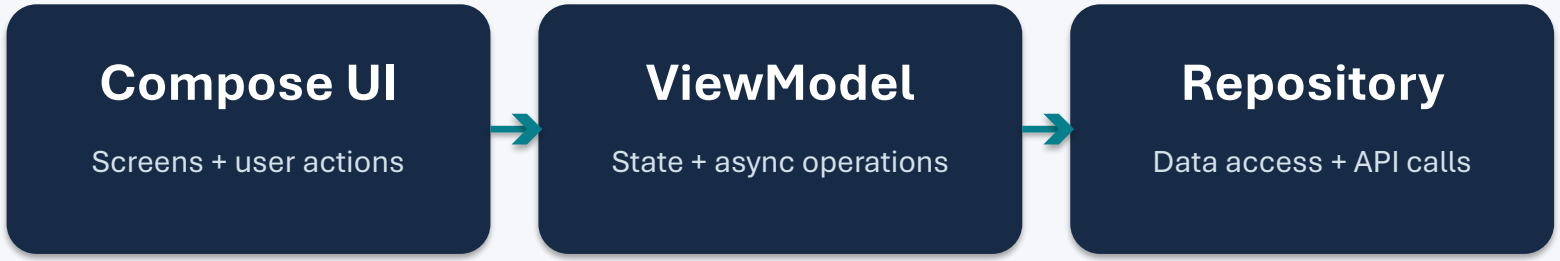
Why MVVM?

UI renders screen state.

ViewModel owns state and asynchronous work.

Repository coordinates local and remote data.

Room emits saved changes.



Two data paths

Local: Room database + DataStore preferences

Remote: Retrofit → Groq API → parsed response

AI guidance becomes a saved study task

1

Set preferences

Company, time budget and preparation topics

2

Call Groq API

Send a request using Retrofit / OkHttp

3

Review output

Parse JSON into weeks and study tasks

4

Save tasks

Convert the plan; store tasks in Room

Local storage

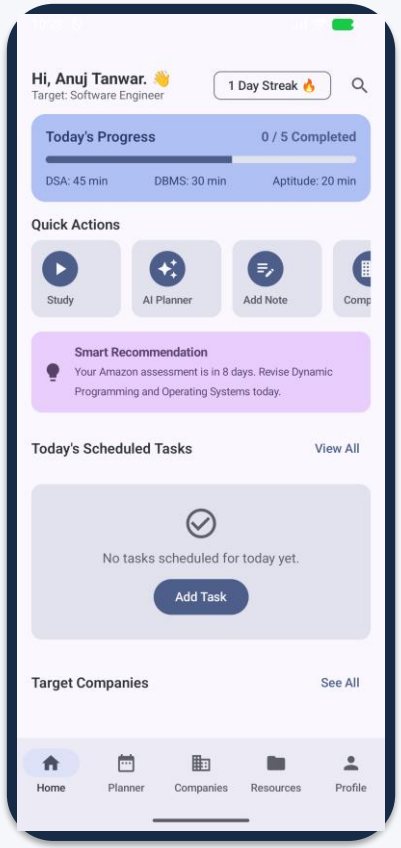
8 Room entities: user, company, question, task, note, experience, resource and bookmark.

Response handling

Loading / success / error UI states.

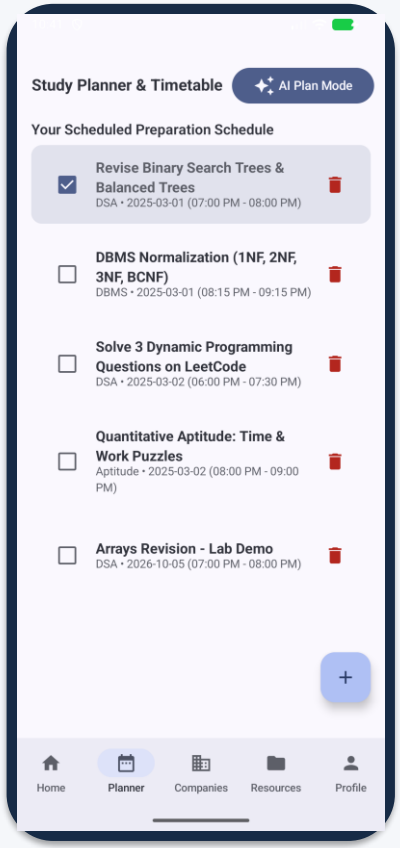
Saved records work offline; AI needs internet.

Important screens in the working application



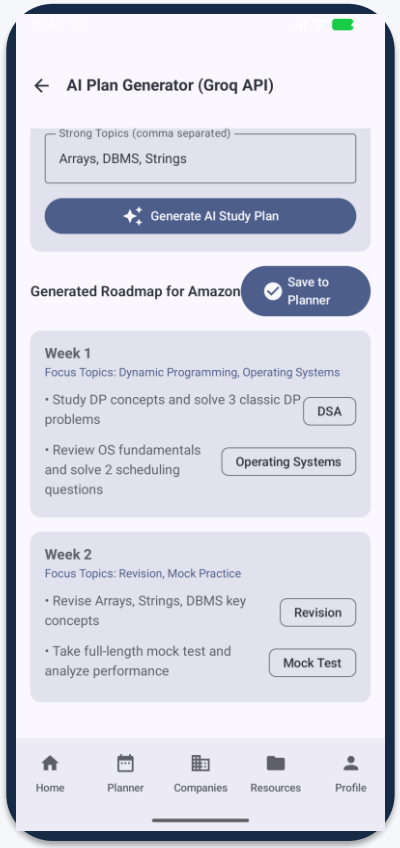
Home

Quick actions & progress



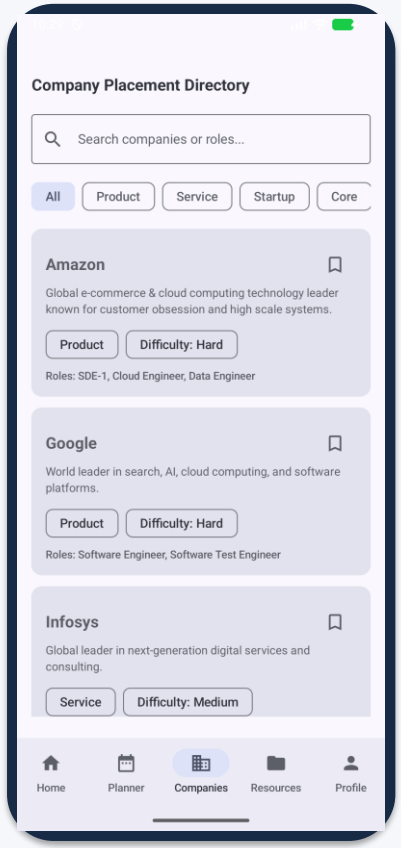
Study planner

Fresh task entry



AI roadmap

Real API output



Companies

Search & company cards

Focused checks verify navigation and live API calls

5 + 3

Navigation checks + live API checks

Navigation: Home returns correctly from planner and other tabs.

API: doubt reply, structured study plan and interview feedback received.

Validation: empty tasks / answers guarded

Challenges resolved

Build compatibility

Aligned Kotlin / Room versions and Gradle plugins.

Unavailable AI model

Updated the model; verified all three API functions.

Home returning to Planner

Corrected saved navigation-state restoration.

Team delivery and individual explanation

Anuj Tanwar | 205124017

Suggested scope: Compose UI, onboarding, Home, planner screens and navigation.

Explain the user journey and lead the UI demo.

Kratish Mewada | 205124045

Suggested scope: Room / DAO, ViewModels, repositories, Groq API and error handling.

Explain data flow and technical verification.

Shared team contribution

Integration, review, testing, demo preparation and overall project understanding.

A practical foundation for placement preparation

Next improvements

Stronger date, time and numeric validation.

More accurate AI task scheduling and real progress analytics.

Backend-managed AI credentials, authentication and cloud sync.

Improved reminders, accessibility and broader device testing.

Project takeaway

Study planning, local storage and AI guidance in one Android app for students.

Thank you • Application demo & individual viva